12500 TI Boulevard, MS 8640, Dallas, Texas 75243

# PCN# 20260330000.1
# Nx556DR and NE556N Design Change

## Change Notification / Sample Request

**Date:** March 31, 2026
**To:** ADVANTECH TECHNOLOGY(CHINA)CO., LTD PCN

Dear Customer:

This is an announcement of a change to a device that is currently offered by Texas Instruments. The details of this change are on the following pages.

Texas Instruments requires acknowledgement of receipt of this notification within 60 days of the date of this notice. Lack of acknowledgement of this notice within 60 days constitutes acceptance and approval of this change. If samples or additional data are required, requests must be received within 60 days of this notification.

The changes discussed within this PCN will not take effect any earlier than the proposed first ship date on Page 3 of this notification, unless customer agreement has been reached on an earlier implementation of the change.

This notice does not change the end-of-life status of any product. Should product affected be on a previously issued product withdrawal/discontinuance notice, this notification does not extend the life of that product or change the life time buy offering/discontinuance plan.

For questions regarding this notice or to provide acknowledgement of this PCN, you may contact your local Field Sales Representative or the change management team.

For sample requests or sample related questions, contact your local Field Sales Representative.

TI values customer engagement and feedback related to TI changes. Customers should contact TI if there are questions or concerns regarding a change notification.

Change Management Team
SC Business Services

**20260330000.1**
**Attachment: 1**

## Products Affected:

The devices listed on this page are a subset of the complete list of affected devices. According to our records, you have recently purchased these devices. The corresponding customer part number is also listed, if available.

| DEVICE | CUSTOMER PART NUMBER |
|---|---|
| NE556N | NULL |
| NA556DR | NULL |

Technical details of this Product Change follow on the next page(s).

| **PCN Number:** | 20260330000.1 | **PCN Date:** | March 31, 2026 |
|---|---|---|---|
| **Title:** | Nx556DR and NE556N Design Change | | |
| **Customer Contact:** | Change Management Team | **Dept:** | Quality Services |
| **Proposed 1st Ship Date:** | June 29, 2026 | **Sample requests accepted until:** | May 30, 2026* |

***Sample requests received after May 30, 2026 will not be supported.**

**Change Type:**

| | | | | | |
|---|---|---|---|---|---|
| ☐ | Assembly Site | ☒ | Design | ☐ | Wafer Bump Material |
| ☐ | Assembly Process | ☐ | Data Sheet | ☐ | Wafer Bump Process |
| ☐ | Assembly Materials | ☐ | Part number change | ☐ | Wafer Fab Site |
| ☐ | Mechanical Specification | ☐ | Test Site | ☐ | Wafer Fab Material |
| ☒ | Packing/Shipping/Labeling | ☐ | Test Process | ☐ | Wafer Fab Process |

## PCN Details

**Description of Change:**

Texas Instruments Incorporated is announcing a design change for Nx556DR and NE556N. The change includes a die redesign to address customer concerns on previous changes made to the 556 Timers.

Qual details are provided in the Qual Data Section.

**Reason for Change:**

Redesign the device to memic the behavior of the legacy device.

**Anticipated impact on Form, Fit, Function, Quality or Reliability (positive / negative):**

Review the SDP and datasheet for full evaluation of the change based on the customer use case.

**Impact on Environmental Ratings:**

Checked boxes indicate the status of environmental ratings following implementation of this change. If below boxes are checked, there are no changes to the associated environmental ratings.

| RoHS | REACH | Green Status | IEC 62474 |
|---|---|---|---|
| ☒ No Change | ☒ No Change | ☒ No Change | ☒ No Change |

**Changes to product identification resulting from this PCN:**

| **Current** Die Rev [2P] | **New Die Rev [2P]** |
|---|---|
| A, B | **B** |

Sample product shipping label (not actual product label):

**Product Affected:**

NA556DR NE556DR NE556N

For alternate parts with similar or improved performance, please visit the product page on TI.com

**Qualification Results**

**Data Displayed as: Number of lots / Total sample size / Total failed**

| Type | # | Test Name | Condition | Duration | Qual Device: NA556DR | Qual Device: NE556DR | QBS Package Reference: LM2901BQDRQ1 | QBS Process Reference: MC33063ADR | QBS Package Reference: OPA4992QDRQ1 | QBS Package Reference: LM2901QDRQ1 |
|---|---|---|---|---|---|---|---|---|---|---|
| HAST | A2 | Biased HAST | 110C/85%RH | 264 Hours | - | - | - | - | 1/77/1[1] | - |
| HAST | A2 | Biased HAST | 130C/85%RH | 96 Hours | - | - | 1/77/0 | - | - | 1/77/0 |
| UHAST | A3 | Unbiased HAST | 130C/85%RH | 96 Hours | - | - | 1/77/0 | - | 1/77/0 | 1/77/0 |
| TC | A4 | Temperature Cycle | -65C/150C | 500 Cycles | - | - | 1/77/0 | - | 1/77/0 | 1/77/0 |
| HTSL | A6 | High Temperature Storage Life | 150C | 1000 Hours | - | - | - | - | 1/77/0 | - |
| HTSL | A6 | High Temperature Storage Life | 175C | 500 Hours | - | - | 1/77/0 | - | 1/77/0 | 1/77/0 |
| HTOL | B1 | Life Test | 125C | 1000 Hours | - | - | - | 3/231/0 | - | - |
| ELFR | B2 | Early Life Failure Rate | 125C | 48 Hours | - | - | - | 3/2400/0 | - | - |
| SD | C3 | PB Solderability | Precondition w.155C Dry Bake (4 hrs +/- 15 minutes) | - | - | - | 1/15/0 | - | - | 1/15/0 |

| Type | # | Test Name | Condition | Duration | Qual Device: NA556DR | Qual Device: NE556DR | QBS Package Reference: LM2901BQDRQ1 | QBS Process Reference: MC33063ADR | QBS Package Reference: OPA4992QDRQ1 | QBS Package Reference: LM2901QDRQ1 |
|---|---|---|---|---|---|---|---|---|---|---|
| SD | C3 | PB-Free Solderability | Precondition w.155C Dry Bake (4 hrs +/- 15 minutes) | - | - | - | 1/15/0 | - | - | 1/15/0 |
| PD | C4 | Physical Dimensions | Cpk>1.67 | - | - | - | 1/10/0 | - | 1/10/0 | 1/10/0 |
| ESD | E2 | ESD CDM | - | 1500 Volts | 1/3/0 | - | - | - | - | - |
| ESD | E2 | ESD HBM | - | 1000 Volts | 1/3/0 | - | - | - | - | - |
| LU | E4 | Latch-Up | Per JESD78 | - | 1/3/0 | - | - | - | - | - |
| CHAR | E5 | Electrical Characterization | Per Datasheet Parameters | - | 1/30/0 | 1/30/0 | - | - | - | - |

- QBS: Qual By Similarity, also known as Generic Data
- Qual Device NA556DR is qualified at MSL1 260C
- Qual Device NE556DR is qualified at MSL1 260C

- Preconditioning was performed for Autoclave, Unbiased HAST, THB/Biased HAST, Temperature Cycle, Thermal Shock, and HTSL, as applicable
- The following are equivalent HTOL options based on an activation energy of 0.7eV : 125C/1k Hours, 140C/480 Hours, 150C/300 Hours, and 155C/240 Hours
- The following are equivalent HTSL options based on an activation energy of 0.7eV : 150C/1k Hours, and 170C/420 Hours
- The following are equivalent Temp Cycle options per JESD47 : -55C/125C/700 Cycles and -65C/150C/500 Cycles

Quality and Environmental data is available at TI's external Web site: http://www.ti.com/

TI Qualification ID: R-CHG-2509-075

**Qualification Results**

**Data Displayed as: Number of lots / Total sample size / Total failed**

| Attributes | Qual Device: NE556N | QBS Package Reference: NE5532P | QBS Process Reference: MC33063ADR | QBS Package Reference: SN74HC00N | QBS Package Reference: ULN2003AN | QBS Package Reference: NE556N | QBS Product Reference: NA556DR |
|---|---|---|---|---|---|---|---|
| **Die Attributes** | | | | | | | |
| **Wafer Fab Supplier** | RFAB | SH-BIP-1 | RFAB | RFAB | RFAB | RFAB | RFAB |
| **Wafer Process** | TIB | JI1 | TIB | LBC9 | TIB | TIB | TIB |
| **Die Size (L,W) (um)** | 988 x 1286 | 2166.62 x 2113.28 | 1563 x 1139 | 460 x 510 | 1068 x 1100 | 1038 x 1336 | 988 x 1286 |
| **Passivation** | Silicon Oxynitride | Silicon Oxynitride | Silicon Oxynitride | Silicon Oxynitride | Silicon Oxynitride | Silicon Oxynitride | Silicon Oxynitride |
| **Package Attributes** | | | | | | | |
| **Assembly Site** | FMX | FMX | MLA | FMX | FMX | FMX | MLA |
| **Package Group** | PDIP | PDIP | SOIC | PDIP | PDIP | PDIP | SOIC |
| **Package Designator** | N | P | D | N | N | N | D |
| **Package Size (mm)** | 19.3 x 6.35 | 9.81 x 6.35 | 4.9 x 3.9 | 19.3 x 6.35 | 19.3 x 6.35 | 19.3 x 6.35 | 8.65 x 3.9 |
| **Body Thickness (mm)** | 3.9 | 3.9 | 1.58 | 3.9 | 3.9 | 3.9 | 1.58 |
| **Pin Count** | 14 | 8 | 8 | 14 | 16 | 14 | 14 |
| **Lead Finish** | NIPDAU | NIPDAU | NIPDAU | NIPDAU | NIPDAU | NIPDAU | NIPDAU |

| Attributes | Qual Device: NE556N | QBS Package Reference: NE5532P | QBS Process Reference: MC33063ADR | QBS Package Reference: SN74HC00N | QBS Package Reference: ULN2003AN | QBS Package Reference: NE556N | QBS Product Reference: NA556DR |
|---|---|---|---|---|---|---|---|
| **Lead Pitch(mm)** | 2.54 | 2.54 | 1.27 | 2.54 | 2.54 | 2.54 | 1.27 |
| **Mount Compound Supplier** | HENKEL | HENKEL | HENKEL | HENKEL | HENKEL | HENKEL | HENKEL |
| **Mount Compound Supplier Number** | QMI 505MT | QMI 505MT | QMI 505MT | QMI 505MT | QMI 505MT | QMI 505MT | QMI 505MT |
| **Mold Compound Supplier** | SUMITOMO | SUMITOMO | SUMITOMO | SUMITOMO | SUMITOMO | SUMITOMO | SUMITOMO |
| **Mold Compound Supplier Number** | EME-G633C | EME-G633C | EME-G633C | EME-G633C | EME-G633C | EME-G633C | EME-G633C |
| **Bond Wire Composition** | CU | CU | CU | CU | CU | CU | CU |
| **Bond Wire Diameter(um)** | 20.32 | 24.384 | 24.384 | 20.32 | 25.4 | 20.3 | 20.32 |
| **Flammability Rating** | UL 94 V-0 | UL 94 V-0 | UL 94 V-0 | UL 94 V-0 | UL 94 V-0 | UL 94 V-0 | UL 94 V-0 |

- QBS: Qual By Similarity, also known as Generic Data
- Qual Device NE556N is qualified at NOT CLASSIFIED NOT CLASSIFIED

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Quality and Environmental data is available at TI's external Web site: http://www.ti.com/

TI Qualification ID: R-CHG-2509-076

In performing change qualifications, Texas Instruments follows integrated circuit industry standards in performing defect mechanism analysis and failure mechanism-based accelerated environmental testing to ensure wafer fab process, assembly process and product quality and reliability. As encouraged by these standards, TI uses both product-specific and generic (family) data in qualifying its changes. For devices to be categorized as a 'product qualification family' for generic data purposes, they must share similar product, wafer fab process and assembly process elements. The applicability of generic data (also known at TI as Qualification by Similarity (QBS)) is determined by the Reliability Engineering function following these industry standards. Generic data is shown in the qualification report in columns titled "QBS Process" (for wafer fab process), "QBS Package" (for assembly process) and "QBS Product" (for product family).

For questions regarding this notice, e-mails can be sent to the Change Management team or your local Field Sales Representative.

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